DEVINE x howest/RESEARCH

# Blender 3D Icon Guidelines

## How to Create & Export Visuals for Howest Research

This document provides practical instructions for creating and exporting 3D icons in Blender. These visuals are used within the Mobile Infopoint for Innovation project to represent the various clusters, themes, and keywords of Howest Research.

# Introduction

## About the 3D Icons

The 3D icons are a consistent set of simple visuals representing the different clusters, themes and keywords. On the browsing screen and on the hologram, each project receives its own unique visual. The icon of the main cluster is placed at the center, and surrounded it by the icons of related keywords and themes. This allows every project to automatically generate a distinct and meaningful visual.

## Purpose of the Guide

As new clusters, themes, or keywords are added to Howest Research, you may need to create new 3D icons or update existing ones. This guide helps you extend your icon library using the provided Blender file and explains how to export them for usage. The instructions to integrate them into the code are explained in the Readme file.

## Who should use this Guide

This guide is intended for designers with knowledge of Blender who are already familiar with 3D modeling. It is not a tutorial on how to learn Blender from scratch. Instead, it provides instructions on how to use the provided Blender starting file, apply the existing style, materials and presets, and correctly export 3D icons for the Mobile Infopoint.

# Getting started

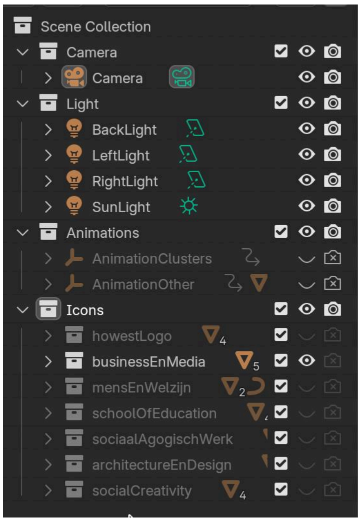
## Needed

- Blender
- Starter file: *'infopointForInnovation_template.blend'*

## File Structure

The provided starter file consists of the following collections:

- **Camera** *(no need to touch)*
  - Camera
- **Light** *(no need to touch)*
  - BackLight
  - LeftLight
  - RightLight
  - SunLight
- **Animations** *(will be used)*
  - AnimationClusters
  - AnimationOther
- **Icons** *(work within this folder)*
  - #IconName




## Get started

To get started, create a new collection within the 'Icons' collection.

Name this new collection according to the cluster, theme, or keyword you will be creating. Use camelCase for the name, and ensure it matches the name used in the coding files.

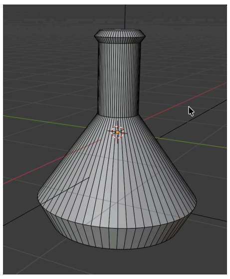
Within this collection, you can begin modelling your icon. To help keep everything visually consistent, please take a look at the style guidelines on the next page. You'll also find several reference icons in this folder, including the Howest logo, five clusters (one of each domain), and a keyword/theme.

The camera, light and animations are already set up for later, no need to adjust these.

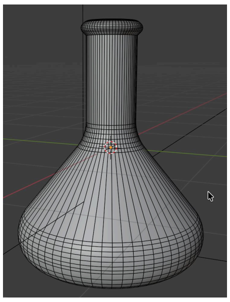
# Style Guidelines

You can now model your icon according to the style guidelines below. You don't need to worry about scale, organization within the folder, or export settings yet.

## Geometry & Complexity

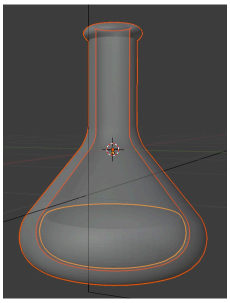


- **Simple:** Icons will often be displayed alongside others, so avoid excessive detail. Keep shapes minimal and avoid using too many separate elements.
- **Thickness**: Avoid very thin or intricate features. Even flat icons should have enough thickness to remain readable and maintain their shape.
- **Gaps**: If the icon consists multiple meshes, it's okay to have certain parts 'floating'. What matters is that the end composition looks good together.
- **Silhouette**: Avoid shapes that are extremely long along a single axis. Even if an element is wider than tall, or taller than wide, ideally, the bounding box of your icon should be roughly square.



## Stylisation

- **Not realistic:** Icons do not need to be fully realistic. Some elements can be slightly exaggerated or out of proportion if it helps make the object more recognisable.
- **Rounded Edges**: Bevel all corners and edges to create a soft, smooth look. This helps create an approachable style similar to that of Howest.
- **Color in glass:** If the icon consists mostly of a glass mesh, this mesh can be filled with a smaller, colored version of the same mesh in the middle. This adds contrast and makes the icon more visually interesting.



# Materials

All materials needed for the icons are included in the provided Blender file and can be applied to the different elements of your icon. Each icon should use a combination of:

- **Frosted Glass**: Provides a soft, slightly translucent look that conveys an approachable yet innovative feel.
- **Matte Color**: Related to the main domains, the matte colors add solidity and visual clarity, complementing the frosted glass elements.

**Material List:**

- Colors Keywords & Themes
- Color Blue - Digitaal/Overschrijdend - Cluster
- Color Green - Ecologsich - Cluster
- Color Pink - Gezond - Cluster
- Color Purple - Sociaal - Cluster
- Color Yellow - Leren - Cluster
- Frosted Glass


### Color for Keywords & Themes

Since these icons need to be rendered in all 5 colors, you can just assign the material ***'Colors Keywords & Themes'*** to the parts where you want color. This material has keyframes that automatically change its color depending on the frame, which will be used later on when rendering


### Color for Clusters

Each cluster is assigned to a main domain with a designated color. When creating a new cluster, check which domain it belongs to and assign the corresponding material to the parts you want colored.

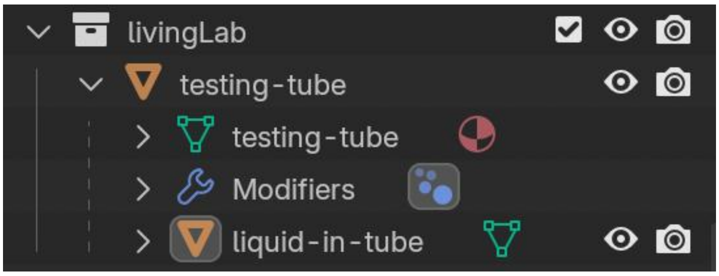
# Preparing for Export

After the icon is modelled and materials are applied, we can prepare the item for export.

## Parenting

- To make it easier to move, rotate, and scale your icon, **parent** all smaller meshes to the largest or central mesh. (CMD + P)
- This ensures all parts stay together when transforming the icon as a single object.

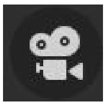
Your structure should look something like this:




## Apply Transformations

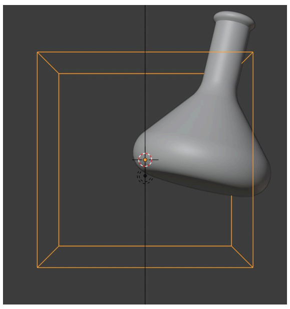
1. **Preview the Render**

- Click the camera icon in Blender to see how the icon will appear when rendered.



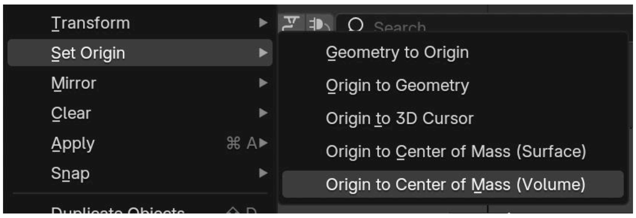
2. **Enable Bounding Box Reference**

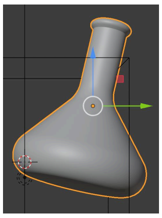
- **For keywords & themes**: Turn on the visibility of the ‘AnimationOther’ object in the ‘Animations’ collection.
- **For clusters**: Turn on the visibility of the ‘AnimationCluster’ object in th ‘Animations’ collection.
- These null objects have a bounding box of 1m × 1m × 1m to guide scaling and positioning. origin of the parent object to the geometric center of the icon. This makes transformations easier and more accurate.



### 3. Set the Origin

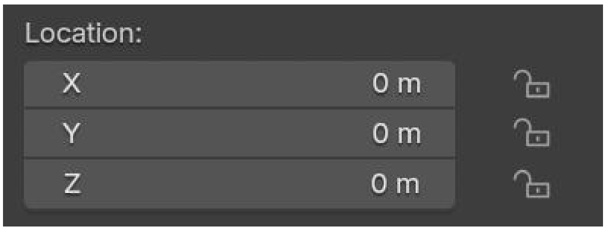
- Set the origin of the parent object to the geometric center of the icon. This makes transformations easier and more accurate.

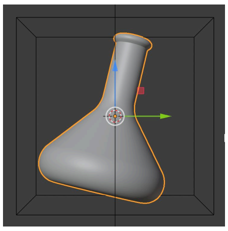




### 4. Position the Icon

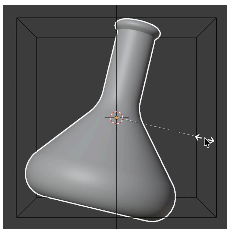
- Move the parent object to the center of the bounding box (Location: 0m, 0m, 0m).
- If the icon doesn't appear fully centered, adjust manually to ensure visual balance.





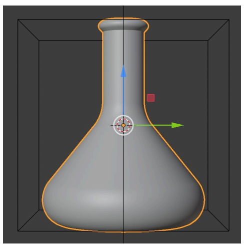
### 5. Scale the Icon

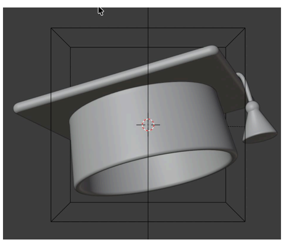
- Scale the parent object so the icon fills the bounding box appropriately.
- Icons that are longer than they are wide (or vice versa) can slightly exceed the box to maintain visual weight consistent with other icons.



### 6. Rotate the Icon

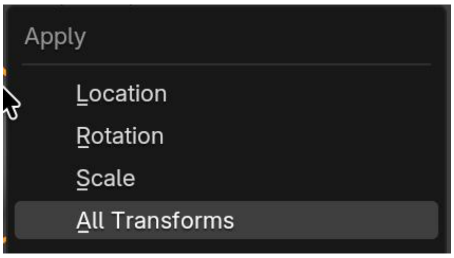
- Rotate the parent object so the front of the icon faces forward.
- **For keywords & themes:** Make sure no extra rotation is applied.
- **For clusters**: Apply the rotation, so it's set to (0 , 0m, 0m). (Cmd + A / Ctrl + A → Rotation). You can now add a slight rotation to the object so it looks the best possible! Since it doesn't have set rotation frames, you can just make sure the main view of this object looks good


(keywords/ themes)


(cluster)

**7. Apply Transformations**

- When the icon is properly positioned, scaled, and rotated, apply all transformations on the parent object. CMD + A / CTRL + A → All transforms
- This ensures a correct animation in the next step.

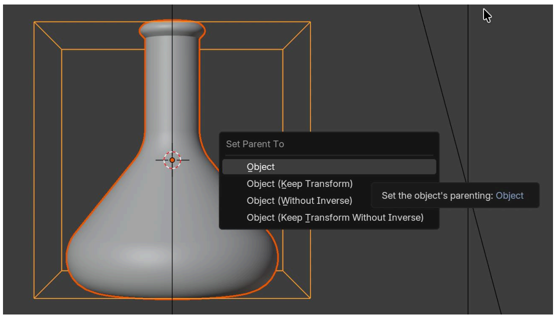


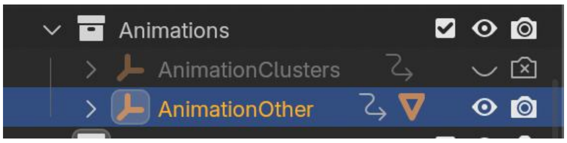

# Link Keyframes

## For keywords & themes

Since we need to render our keywords and themes in different colors and rotations, we will link the parent object to a null object containing all of the keyframes for the different rotations. The keyframes for the different colors are already applied to the *'Colors Keywords & Themes'* material

- Parent the parent object to the '*AnimationOther*' box. (CMD + P)

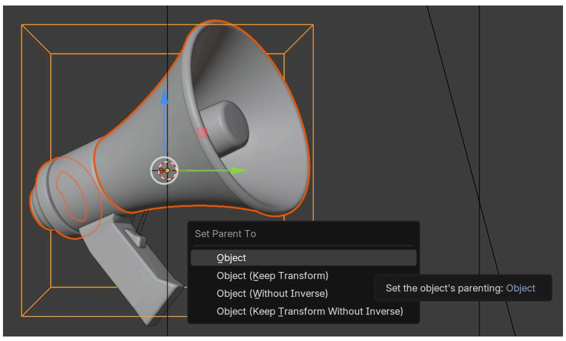


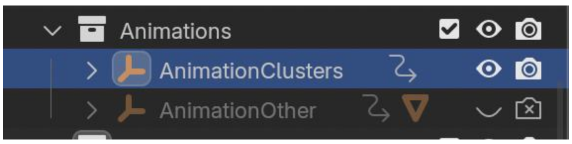




## For clusters

Since we need to render an animation for the clusters, we will link the parent object of our icon to a null object containing all the keyframes for this animation.

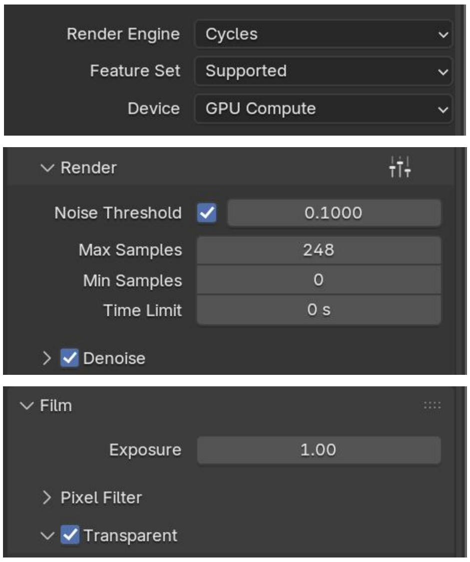
- Parent the parent object to the *'AnimationCluster'* box. (CMD + P)



# Exporting

## Render Settings

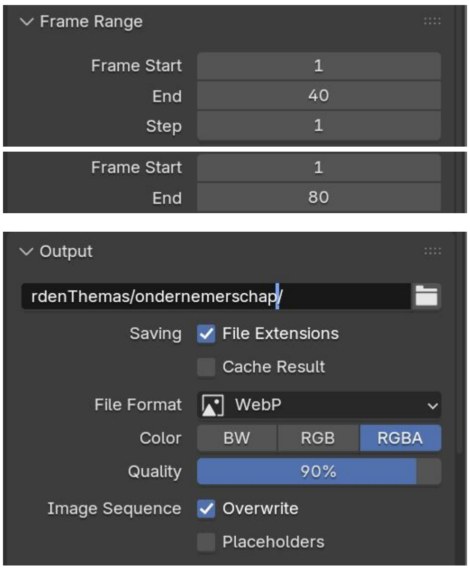



1. **General**
   - Set **Engine** to Cycles
   - If possible, use GPU for faster rendering

2. **Sampling → Render**
   - **Noise Threshold**: 0.1
   - **Max Samples**: min 248

3. **Film**
   - Xheck the **Transparent** box

## Output Settings




1. **Frame Range**
   - Frame **start**: 1
   - Frame **end**:
     - for keywords & themes: 40
     - for clusters: 80

2. **Output**
   - **Folder**: Same name as the collection name (see page 3)
   - Check the **File Extensions** box
   - **File Format**: WebP
   - **Color**: RGBA

## Rendering

Finally, render the icon as an image sequence. Choose the **Render Animation** option so Blender outputs a series of WebP frames. These frames will be handled and combined automatically in the code, so you don't need to manage or edit the sequence yourself. The instructions to correctly add them are provided in the Readme file of the code